# General solution $y(t) = y_c(t) + y_p(t)$ of $y'(t) = Ay(t) + f(t)$

- The **Complementary Solution** satisfies $\boldsymbol{y}_c' = \boldsymbol{A}\boldsymbol{y}_c$ with an eigen-basis $\boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$

$$\boldsymbol{y}_c(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t}\boldsymbol{v}_i = \underbrace{\begin{pmatrix}\boldsymbol{v}_1 & \cdots & \boldsymbol{v}_n\end{pmatrix}}_{V} \begin{pmatrix} e^{\lambda_1 t} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n t}\end{pmatrix} \underbrace{\begin{pmatrix} c_1 \\ \vdots \\ c_n\end{pmatrix}}_{\boldsymbol{c}}$$

- A **Particular Solution** satisfies $\boldsymbol{y}_p' = \boldsymbol{A}\boldsymbol{y}_p + \boldsymbol{f}(t)$ with determined coefficients.
  - If $\boldsymbol{f}(t)$ is polynomial guess $\boldsymbol{y}_p(t) = \boldsymbol{a}_0 + \boldsymbol{a}_1 t + \ldots$
  - If $\boldsymbol{f}(t)$ is exponential guess $\boldsymbol{y}_p(t) = \boldsymbol{a}e^{\alpha t}$ with matching $\alpha$.
  - If $\boldsymbol{f}(t)$ is cos and/or sin guess $\boldsymbol{y}_p(t) = \boldsymbol{a}\cos(\omega t) + \boldsymbol{b}\sin(\omega t)$ with matching $\omega$
  - Undetermined coefficients $\boldsymbol{a}_i$ etc are determined by plugging into $\boldsymbol{y}_p' = \boldsymbol{A}\boldsymbol{y}_p + \boldsymbol{f}(t)$
- Solving $V\boldsymbol{c} = \boldsymbol{y}_0 - \boldsymbol{y}_p(0)$ gives $\boldsymbol{c}$ satisfying the **IC** $\boldsymbol{y}(0) = \boldsymbol{y}_c(0) + \boldsymbol{y}_p(0) = \boldsymbol{y}_0$.

## Complex Eigenpairs $y' = \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix} y = Ay$

- $\boxed{\lambda = -1 \pm 2i}$ since $\det(\boldsymbol{A} - \lambda I) = (\lambda + 1)^2 + 4 = 0 \implies (\lambda + 1)^2 = -4$
- $\boxed{\lambda_1 = -1 + 2i}$ $\boxed{\boldsymbol{v}_1 = (1, -i)}$ since $(\boldsymbol{A} - \lambda_1 I)\boldsymbol{v}_1 = \boldsymbol{0}$

$$\boldsymbol{A} - (-1 + 2i)I = \begin{pmatrix} -2i & -2 \\ 2 & -2i \end{pmatrix} \qquad \left[\begin{array}{cc|c} -2i & -2 & 0 \\ 2 & -2i & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -i & 0 \\ 0 & 0 & 0 \end{array}\right]$$

- $\boxed{\lambda_2 = -1 - 2i}$ $\boxed{\boldsymbol{v}_2 = (1, +i)}$ since $(\boldsymbol{A} - \lambda_2 I)\boldsymbol{v}_2 = \boldsymbol{0}$

$$\boldsymbol{A} - (-1 - 2i)I = \begin{pmatrix} +2i & -2 \\ 2 & +2i \end{pmatrix} \qquad \left[\begin{array}{cc|c} +2i & -2 & 0 \\ 2 & +2i & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & +i & 0 \\ 0 & 0 & 0 \end{array}\right]$$

- For $c_1, c_2 \in \mathbb{C}$ the general solution is

$$\boxed{\boldsymbol{y}(t) = c_1 e^{(-1+2i)t} \begin{pmatrix} 1 \\ -i \end{pmatrix} + c_2 e^{(-1-2i)t} \begin{pmatrix} 1 \\ +i \end{pmatrix}}$$

# Complex Exponentials $e^{(\alpha \pm \beta i t)} = e^{\alpha t} e^{\pm \beta i t}$

Maclaurin/Taylor series

$$\begin{aligned} e^x &= 1 + 1x + (1/2!)x^2 + (1/3!)x^3 + (1/4!)x^4 + \cdots \\ \cos(x) &= 1 + 0x - (1/2!)x^2 + 0x^3 + (1/4!)x^4 + \cdots \\ \sin(x) &= 0 + 1x + 0x^2 - (1/3!)x^3 + 0x^4 + \cdots \end{aligned}$$

$$\begin{aligned} e^{iy} &= 1 + iy - (1/2!)y^2 - (i/3!)y^3 + (1/4!)y^4 + \cdots \\ \cos(y) &= 1 + 0y - (1/2!)y^2 + 0y^3 + (1/4!)y^4 + \cdots \\ i\sin(y) &= 0 + iy + 0y^2 - (i/3!)y^3 + 0y^4 + \cdots \\ \cos(y) + i\sin(y) &= 1 + iy - (1/2!)y^2 - (i/3!)y^3 + (1/4!)y^4 + \cdots \end{aligned}$$

So $\boxed{e^{iy} = \cos(y) + i\sin(y)}$ and $\boxed{e^{(\alpha \pm \beta i)t} = e^{\alpha t}(\cos(\beta t) \pm i\sin(\beta t))}$

## Real forms for complex eigenpairs $y' = \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix} y = Ay$

- $\boxed{\lambda_\pm = -1 \pm 2i}$ $\boxed{\boldsymbol{v}_+ = \begin{pmatrix} 1 \\ -i \end{pmatrix}}$ $\boxed{\boldsymbol{v}_- = \begin{pmatrix} 1 \\ +i \end{pmatrix}}$ $\boxed{\boldsymbol{v}_\pm = \begin{pmatrix} 1 \\ \mp i \end{pmatrix}}$

$$e^{(-1+2i)t}\boldsymbol{v}_+ = e^{-t}(\cos(2t) + i\sin(2t))\begin{pmatrix} 1 \\ -i \end{pmatrix} = e^{-t}\left[\begin{pmatrix} \cos(2t) \\ \sin(2t) \end{pmatrix} + i\begin{pmatrix} \sin(2t) \\ -\cos(2t) \end{pmatrix}\right]$$

$$e^{(-1-2i)t}\boldsymbol{v}_- = e^{-t}(\cos(2t) - i\sin(2t))\begin{pmatrix} 1 \\ +i \end{pmatrix} = e^{-t}\left[\begin{pmatrix} \cos(2t) \\ \sin(2t) \end{pmatrix} - i\begin{pmatrix} \sin(2t) \\ -\cos(2t) \end{pmatrix}\right]$$

  - Adding gives $\boxed{2e^{-t}\begin{pmatrix} \cos(2t) \\ \sin(2t) \end{pmatrix}}$. Subtracting gives $\boxed{2ie^{-t}\begin{pmatrix} \sin(2t) \\ -\cos(2t) \end{pmatrix}}$

For $c_1, c_2 \in \mathbb{R}$ $\boxed{\boldsymbol{y}(t) = c_1 e^{-t}\begin{pmatrix} \cos(2t) \\ \sin(2t) \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} \sin(2t) \\ -\cos(2t) \end{pmatrix}}$

## Real Form from $\lambda_\pm = \alpha \pm \beta i$ and $\boldsymbol{v}_\pm = \boldsymbol{a} \pm \boldsymbol{b}i$

- Expand $e^{\lambda_+ t}\boldsymbol{v}_+ = e^{\alpha t}\left[(\boldsymbol{a}\cos(\beta t) - \boldsymbol{b}\sin(\beta t)) + i(\boldsymbol{a}\sin(\beta t) + \boldsymbol{b}\cos(\beta t))\right]$.
  Real and imaginary parts give two real solutions

$$\boxed{\boldsymbol{y}_1 = e^{\alpha t}(\boldsymbol{a}\cos(\beta t) - \boldsymbol{b}\sin(\beta t))} \qquad \boxed{\boldsymbol{y}_2 = e^{\alpha t}(\boldsymbol{a}\sin(\beta t) + \boldsymbol{b}\cos(\beta t)}$$

- The real solution form, for $\boldsymbol{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \in \mathbb{R}^2$, is

$$\boldsymbol{y}(t) = \boxed{c_1 e^{\alpha t}(\boldsymbol{a}\cos(\beta t) - \boldsymbol{b}\sin(\beta t)) + c_2 e^{\alpha t}(\boldsymbol{a}\sin(\beta t) + \boldsymbol{b}\cos(\beta t))}$$

  As $t \to \infty$ the ith component $\boxed{\sim e^{\alpha t}\cos(\beta t + \phi_i)}$.

- Solve the linear system $\begin{pmatrix} \boldsymbol{a} & \boldsymbol{b} \end{pmatrix}\boldsymbol{c} = \boldsymbol{y}_0$ to determine $c_1$ and $c_2$ to satisfy $\boldsymbol{y}(0) = \boldsymbol{y}_0$.

## $y' = Ay + f(t)$ Complementary Solution $y_c$

$$\boldsymbol{A} = \begin{pmatrix} 1 & -2 & -2 & 0 \\ -2 & 1 & 4 & 2 \\ -2 & -2 & 1 & 0 \\ -2 & -2 & 0 & -3 \end{pmatrix} \qquad \begin{aligned} \lambda_1 &= -1 + 2i \\ \lambda_2 &= -1 - 2i \\ \lambda_3 &= \phantom{-}3 \\ \lambda_4 &= -1 \end{aligned} \qquad V = \begin{array}{c} \begin{matrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 & \boldsymbol{v}_4 \end{matrix} \\ \begin{pmatrix} 1 & 1 & -1 & -1 \\ -i & +i & 1 & 0 \\ 1 & 1 & 0 & -1 \\ +i & -i & 0 & 1 \end{pmatrix} \end{array}$$

For $c_1, c_2, c_3, c_4 \in \mathbb{R}$ the real GS is

$$\boldsymbol{y}_c(t) = \sum_{k=1}^{4} c_k \boldsymbol{y}_k(t) = c_1 e^{-t} \begin{pmatrix} \cos(2t) \\ \sin(2t) \\ \cos(2t) \\ -\sin(2t) \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} \sin(2t) \\ -\cos(2t) \\ \sin(2t) \\ \cos(2t) \end{pmatrix} + c_3 e^{3t} \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c_4 e^{-t} \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix}$$

For $c_3 \neq 0$ the complementary solution satisfies $\boldsymbol{y}_c(t) \sim c_3 e^{3t} \boldsymbol{v}_3$ as $t \to \infty$.

## $y' = Ay + e^{2t}(1,-1,0,0)^T + \cos(3t)(3,-2,3,2)^T$ **Particular Solution** $y_p = y_{p_1} + y_{p_2}$

- Substitute $y_{p_1} = ae^{2t}$ and $y'_{p_1} = 2ae^{2t}$ in $\boxed{y'_{p_1} = Ay_{p_1} + e^{2t}(1,-1,0,0)^T}$ to get

$$2ae^{2t} = Aae^{2t} + e^{2t}(1,-1,0,0)^T \implies (2I - A)a = (1,-1,0,0)^T \implies a = (-1,1,0,0)^T$$

- Substitute $y_{p_2} = b\cos(3t) + c\sin(3t)$ and $y'_{p_2} = -3b\sin(3t) + 3c\cos(3t)$

$$-3b\sin(3t) + 3c\cos(3t) = (Ab + (3,-2,3,2)^T)\cos(3t) + Ac\sin(3t)$$

$$\begin{aligned} 3c &= Ab + (3,-2,3,2)^T \\ -3b &= Ac \end{aligned} \implies \begin{aligned} b &= (1/2,-1,1/2,1)^T \\ c &= (3/2,0,3/2,0)^T \end{aligned}$$

$$\boxed{y_p(t) = e^{2t}\begin{pmatrix}-1\\1\\0\\0\end{pmatrix} + (1/2)\begin{pmatrix}1\\-2\\1\\2\end{pmatrix}\cos(3t) + (3/2)\begin{pmatrix}1\\0\\1\\0\end{pmatrix}\sin(3t)}$$

## $y' = Ay + f(t)$ with Initial condition $y(0) = y_0 = (-2, 0, 0, 3)^T$

Plug in $t = 0$

$$\boldsymbol{y}_p(0) = (-1, 1, 0, 0)^T + (1/2)(1, -2, 1, 2)^T = (-1/2, 0, 1/2, 1)^T$$

$$\boldsymbol{y}_c(0) = \boldsymbol{y}_0 - \boldsymbol{y}_p(0) = (-2, 0, 0, 3)^T - (-1/2, 0, 1/2, 1)^T = (-3/2, 0, -1/2, 2)^T$$

With $M = [\boldsymbol{y}_1(0)\ \boldsymbol{y}_2(0)\ \boldsymbol{y}_3(0)\ \boldsymbol{y}_4(0)]$ the LS for the IC is $M\boldsymbol{c} = (-3/2, 0, -1/2, 2)^T$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & -1 & -3/2 \\ 0 & -1 & 1 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1/2 \\ 0 & 1 & 0 & 1 & 2 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & 0 & -1 & -1 & -3/2 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right] \Longrightarrow \boldsymbol{c} = \begin{pmatrix} 1/2 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

IVP solution

$$\boldsymbol{y}(t) = \boldsymbol{y}_c(t) + \boldsymbol{y}_p(t) = \sum_{k=1}^{4} c_k \boldsymbol{y}_k(t) + \boldsymbol{y}_{p_1}(t) + \boldsymbol{y}_{p_2}(t)$$

## Summary $y' = Ay + f_1(t) + f_2(t)$ $y = y_c + y_p = \sum_{k=1}^{n} c_k y_k + y_{p_1} + y_{p_2}$

- **Complementary Solution** If $\boldsymbol{A}\boldsymbol{v}_k = \lambda_k \boldsymbol{v}_k$ for $1 \leq k \leq n$ and $M = [\boldsymbol{v}_1 \ \boldsymbol{v}_2 \dots \boldsymbol{v}_n]$ is invertible construct $\boldsymbol{y}_c(t) = \sum_{k=1}^{4} d_k e^{\lambda_t} k \boldsymbol{v}_k$ from $n$ LI real solutions.
  - Real eigenvalue $\lambda$ and eigenvector $\boldsymbol{v}$ give the real solution $\boxed{e^{\lambda t}\boldsymbol{v}}$
  - Conjugate eigenvalues $\lambda = \alpha \pm \beta i$ and eigenvectors $\boldsymbol{v} = \boldsymbol{a} \pm \boldsymbol{b}i$ give two real solutions $\boxed{e^{\alpha t}(\boldsymbol{a}\cos(\beta t) - \boldsymbol{b}\sin(\beta t))}$ and $\boxed{e^{\alpha t}(\boldsymbol{a}\sin(\beta t) + \boldsymbol{b}\cos(\beta t))}$
- **Particular solution** construct $\boldsymbol{y}_p(t) = \boldsymbol{y}_{p_1}(t) + \boldsymbol{y}_{p_2}(t)$ where $\boldsymbol{y}'_{p_j}(t) = \boldsymbol{A}\boldsymbol{y}_{p_j}(t) + \boldsymbol{f}_j(t)$. Match undetermined coefficients in $\boldsymbol{a}_0 + \boldsymbol{a}_1 t + \cdots$ for a polynomial in $\boldsymbol{a}e^{\gamma t}$ for an exponential, and $\boldsymbol{a}\cos(\omega t) + \boldsymbol{b}\sin(\omega t)$ for a sinusoid.
- **IC** Solve $M\boldsymbol{c} = \boldsymbol{y}_0 - \boldsymbol{y}_p(0)$ with $\boldsymbol{M} = \begin{pmatrix} \boldsymbol{y}_1(0) & \cdots & \boldsymbol{y}_n(0) \end{pmatrix}$ to enforce IC $\boldsymbol{y}(0) = \boldsymbol{y}_0$.